BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Ethical Analysis Report

Group ID: 25-26J-123
Project Title: AUTOSDC: Agentic AI for Automatic SDC Pipeline Selection in Census Microdata

Ethical Dimension	Identification and Analysis
Data Privacy Concerns	AUTOSDC processes highly sensitive census microdata sourced from the Department of Census and Statistics (DCS) of Sri Lanka — including household-level attributes such as age, sex, district, income bracket, religion, and health condition — which, in combination, constitute a significant re-identification risk. The following privacy concerns have been identified and formally addressed within the scope of this research: **1. Sensitivity of Census Microdata:** The 2012 Sri Lankan Population and Housing Census microdata sample (85,432 records) contains quasi-identifiers such as district, age group, sex, religion, and marital status that could enable a motivated adversary to re-identify specific individuals via record linkage with publicly available registers. AUTOSDC directly addresses this by applying k-anonymity ($k \geq 5$), l-diversity ($l \geq 3$), and t-closeness ($t \leq 0.25$) transformations so that no individual record is distinguishable from at least $k-1$ others with respect to all quasi-identifier attributes. **2. Informed Consent and Legal Basis:** Census data was collected under the legal authority of the Statistics Ordinance of Sri Lanka, which mandates confidential handling of individual responses. The research team obtained formal written permission from DCS to use a de-identified microdata sample exclusively for academic research. No additional primary data collection from human participants was conducted. The research complies with the Sri Lanka Data Protection Act (2022) and SLIIT Research Ethics Guidelines, ensuring that no personal data is stored beyond the research period and that all outputs undergo SDC verification prior to dissemination. **3. Residual Disclosure Risk:** AUTOSDC acknowledges that no SDC method can guarantee absolute zero disclosure risk. Therefore, the system employs a multi-layered evaluation combining formal privacy-model verification with record linkage simulation (IHMT method) to estimate residual re-identification probability for each candidate anonymized dataset. Only pipelines whose residual risk falls below DCS-mandated thresholds are approved for release. **4. Attribute Disclosure Risk:** Beyond identity disclosure, the system guards against inference attacks in which an adversary deduces sensitive attributes (e.g., income bracket, health condition) without directly identifying an individual. The l-diversity constraint ensures heterogeneity of sensitive values within each equivalence class, while t-closeness ensures that the distribution of sensitive attributes does not deviate significantly from the overall population distribution, preventing distributional inference attacks.

Data Security	Given that AUTOSDC handles raw, unprotected census microdata during its processing pipeline, robust technical and operational security measures are essential to prevent unauthorized access, data breaches, or inadvertent exposure. The following data security controls are implemented across all phases of the research: **1. Data Access Controls and Isolation:** Raw microdata provided by DCS is stored exclusively on a password-protected, air-gapped research workstation located within SLIIT premises. Access is restricted to the principal researcher (Madhuwarsha Thilagaratnam) and the research supervisor (Dr. Prasanna Sumathipala). No raw microdata is transmitted over public networks, uploaded to cloud services, or stored on removable media without prior encryption. Role-based access controls (RBAC) are enforced at the operating system level, and all access attempts are logged and auditable. **2. Encryption at Rest and in Transit:** All microdata files and intermediate processing outputs are encrypted at rest using AES-256 encryption. Any necessary data transfer between research environments (e.g., for supervisor review) is conducted via TLS 1.3-secured channels or encrypted removable media. Encryption keys are managed using a hardware-backed key management approach, and no unencrypted copies of raw microdata are retained beyond the active processing session. **3. Containerized and Isolated Processing Environment:** AUTOSDC's processing pipeline is deployed within Docker containers with no external network access during active processing. This containerized architecture ensures that any vulnerability in a third-party library cannot be exploited to exfiltrate microdata, and that the processing environment is reproducible, isolated, and auditable. **4. Data Minimization and Retention Policy:** In compliance with the data minimization principle, only the variables strictly necessary for research objectives (seven quasi-identifiers and three sensitive attributes) are retained during processing; all other variables are stripped from the working dataset before analysis commences. All raw microdata and intermediate non-anonymized files will be securely deleted upon project completion, with deletion verified and documented in accordance with DCS data governance requirements. **5. Dependency and Supply-Chain Security:** All third-party Python and R libraries used by AUTOSDC are version-pinned in requirements files and sourced exclusively from official package repositories (PyPI, CRAN). A software bill of materials (SBOM) is maintained, and dependency versions are periodically reviewed for known security vulnerabilities (CVEs) using automated scanning tools. Any vulnerable dependency is patched or replaced before deployment.

Bias and Fairness	AUTOSDC operates on census data that reflects the demographic and socioeconomic composition of Sri Lanka's population. SDC methods, if applied without careful consideration of fairness, can disproportionately degrade data quality for minority or underrepresented subgroups, thereby introducing or amplifying biases that may affect downstream statistical analyses and policy decisions. The following bias and fairness concerns have been identified and systematically addressed: **1. Disproportionate Suppression of Minority Subgroups:** Local suppression and global recoding techniques can disproportionately affect records belonging to small demographic subgroups (e.g., minority ethnic groups or residents of sparsely populated districts) who are more likely to form unique combinations of quasi-identifier values. AUTOSDC mitigates this by incorporating a fairness-aware suppression constraint within the NSGA-II objective function, limiting the suppression rate for any demographic subgroup to within 1.5× the overall dataset suppression rate. This ensures that the burden of anonymization does not fall disproportionately on already-marginalized populations. **2. Knowledge Base Bias:** The AUTOSDC rule-based knowledge base was constructed through structured expert elicitation from SDC specialists. There is an inherent risk that expert heuristics may encode institutional biases or assumptions that are not appropriate for all subpopulations in Sri Lankan census data. To mitigate this, the knowledge base was reviewed by two independent domain experts, and all applicability rules were validated against representative stratified samples of the dataset to ensure they do not systematically disadvantage any demographic group. **3. Utility Preservation Fairness:** AUTOSDC evaluates information loss not only at the aggregate dataset level but also at the subgroup level, using stratified information loss metrics. The NSGA-II optimization includes a subgroup utility preservation objective that penalizes solutions exhibiting high information loss variance across demographic strata. This ensures that the analytical value of the released data is preserved equitably across all population groups, supporting fair evidence-based policy-making. **4. LLM-Assisted Classification Bias:** AUTOSDC employs a fine-tuned large language model (LLM) to assist in quasi-identifier detection from variable metadata. LLMs are known to exhibit biases inherited from their training corpora. To address this, the LLM classifier's output is treated as advisory only and is validated against the deterministic rule-based classifier before any variable is classified as a quasi-identifier or sensitive attribute. Disagreements are flagged for

	human expert review, ensuring that no automated misclassification adversely affects the anonymization outcome.
Transparency and Accountability	AUTOSDC is designed with transparency and accountability as first-class design principles. Autonomous AI systems that make consequential decisions — such as selecting and applying privacy-altering transformations to national census data — must be interpretable, auditable, and subject to meaningful human oversight. The following measures ensure these properties throughout the research and deployment lifecycle: **1. Reproducible Anonymization Recipes:** For every anonymized dataset produced by AUTOSDC, the system generates a complete, machine-readable "anonymization recipe" that specifies, in precise terms, every SDC technique applied, the exact parameter values used, the sequence of application, the quasi-identifiers and sensitive attributes targeted, and the privacy and utility metrics achieved at each step. This recipe can be re-executed on the original data to reproduce the identical anonymized output, enabling independent verification and regulatory audit. Empirical evaluation confirmed 100% reproducibility across 20 independent runs. **2. Human-in-the-Loop (HITL) Oversight:** AUTOSDC's agentic workflow incorporates mandatory human review checkpoints at three stages: (a) after quasi-identifier and sensitive attribute classification, (b) after the Pareto front is generated but before a solution is selected for implementation, and (c) after the final anonymized dataset is produced but before it is approved for dissemination. A DCS statistician must explicitly approve each checkpoint before the workflow proceeds, ensuring that no dataset is released without qualified human review. **3. Explainability of AI Decisions:** For each SDC technique recommended by the Recommendation Engine, AUTOSDC provides a natural language explanation of why that technique was selected, referencing the specific data characteristics and knowledge base rules that triggered the recommendation. For the NSGA-II optimizer, the system presents a visual Pareto front and explains the trade-off implications of each candidate solution in plain language accessible to non-specialist DCS staff, supporting informed decision-making. **4. Research Ethics Compliance and Documentation:** This research was conducted in full compliance with SLIIT's research ethics framework. All project documentation, including the data access agreement with DCS, the research ethics declaration, and all anonymization recipes, is maintained in a version-controlled repository accessible to the research supervisor. The research team accepts full accountability for the methodological choices made in AUTOSDC and for ensuring that the system does not produce outputs

	that could harm data subjects or undermine public trust in national statistical institutions. **5. Open Methodology and Peer Review:** The AUTOSDC codebase (excluding raw microdata) is published in a public GitHub repository under an open-source MIT licence, enabling independent replication and peer review of the methodology. All evaluation results reported in this dissertation are accompanied by the analysis scripts used to produce them, supporting full reproducibility. The research findings will be submitted for publication in a peer-reviewed venue, subjecting the methodology to formal independent scrutiny.
Impact on Stakeholders	AUTOSDC affects a diverse range of stakeholders whose interests must be carefully balanced. The following analysis evaluates the potential positive and negative impacts on each stakeholder group, and describes the measures taken to ensure that the system operates in their best interests: **1. Census Data Subjects (Sri Lankan Residents):** The most fundamental stakeholders are the 85,432 individuals whose census records are processed by AUTOSDC. These data subjects did not provide consent for research use of their data (census participation is legally mandated); however, they have a legitimate expectation that their personal information will be protected against disclosure. AUTOSDC directly upholds this expectation by transforming their records to satisfy formal privacy guarantees (k-anonymity, l-diversity, t-closeness) before any release. The system's fairness-aware suppression constraints additionally protect minority subgroup members from disproportionate privacy degradation. Data subjects are further protected by the secure processing environment, strict access controls, and the legal framework of the DCS data access agreement. **2. Department of Census and Statistics (DCS):** DCS stands to benefit significantly from AUTOSDC as the primary intended institutional user of the system. The framework addresses DCS's operational challenges directly: it reduces the time required to produce a privacy-certified microdata release from an estimated 180–240 minutes (manual workflow) to 22–31 minutes, substantially reduces the specialist expertise burden on DCS staff, and produces fully reproducible and auditable anonymization recipes that support regulatory compliance reporting. The risk to DCS is minimal: the human-in-the-loop oversight design ensures that DCS statisticians retain final decision-making authority over all data releases, and the system cannot autonomously publish any dataset. **3. Researchers and Policy Analysts (Data Users):** Academic researchers, government analysts, and international development organizations that rely on DCS microdata for policy research benefit

	from AUTOSDC through improved access to higher-quality, more analytically useful anonymized datasets. AUTOSDC's NSGA-II optimization preserves key statistical properties (distributional characteristics, regression coefficients, cross-tabulation frequencies) with an average information loss of only 7.9%, compared to 22–35% for individual manually-tuned techniques. The risk to this stakeholder group is that over-anonymization could degrade the analytical value of data; AUTOSDC explicitly minimizes this risk through its utility-preserving multi-objective optimization framework. **4. DCS Professional Staff (System Operators):** DCS statisticians who currently perform manual SDC are impacted by AUTOSDC in that their roles will shift from hands-on parameter tuning to high-level oversight and review. This transition carries both positive impacts (reduced cognitive burden, elimination of repetitive manual tasks, more time for strategic analytical work) and potential negative concerns (deskilling risk if staff cease to maintain deep SDC expertise, or anxiety about job displacement). AUTOSDC is designed to augment rather than replace human expertise: the HITL oversight model and the system's transparent explainability outputs ensure that staff remain engaged, informed, and professionally empowered throughout the anonymization process. **5. Regulatory and Oversight Bodies:** The Personal Data Protection Authority of Sri Lanka and other regulatory bodies have a direct interest in ensuring that census microdata releases comply with the Sri Lanka Data Protection Act (2022). AUTOSDC supports regulatory compliance by generating comprehensive audit trails and certification reports that document the privacy guarantees achieved for each dataset release. These reports can be submitted directly to oversight bodies as evidence of due diligence, reducing regulatory burden for DCS and enhancing public accountability for data dissemination decisions. The open-source availability of AUTOSDC's codebase further supports independent regulatory audit of the anonymization methodology itself.